SCAMSHIELD AI

Explainable, Multi-Signal Cyber Scam Pattern Recognition & Early-Warning Platform

Target Domain & Hackathon Track
RAKSHAM AI Cybersecurity Hackathon | IIT Delhi
Problem Statement 2: AI-Driven Scam Pattern Recognition

Core Engineering Thesis
Halting multi-stage UPI, banking KYC, utility, and digital arrest scams *before* financial exfiltration occurs—without black-box opacity, high latency, or external cloud lock-in.

1.16 ms	19 Signals	100.0%	52 / 52	0 Cloud APIs
Mean CPU Latency	5 Threat Clusters	F1-Score (N=60 Test)	Hermetic Tests Passing	100% Offline Edge Capable



THE PROBLEM: MULTI-STAGE SOCIAL ENGINEERING IN INDIA

Why Traditional Spam Filters and Cloud LLMs Fail Against Modern Cyber Syndicates

Dominant Indian Cyber Threat Vectors

- **Digital Arrest Extortion**: Coercive fake police/CBI Skype warrants accusing victims of narcotics trafficking.
- **Immediate Utility Cutoff Panic**: Coercive notices threatening power disconnection at 9:30 PM tonight.
- **UPI PIN Inversion Fraud**: Misleading QR codes and collect requests claiming to credit 'instant refunds'.
- **Banking KYC Suspensions**: Spoofed SBI/HDFC alerts demanding Aadhaar/PAN validation on lookalike domains.
- **Malicious APK Distribution**: Delivery of Trojanized applications (e.g., QuickBill.apk) to steal SMS OTPs.

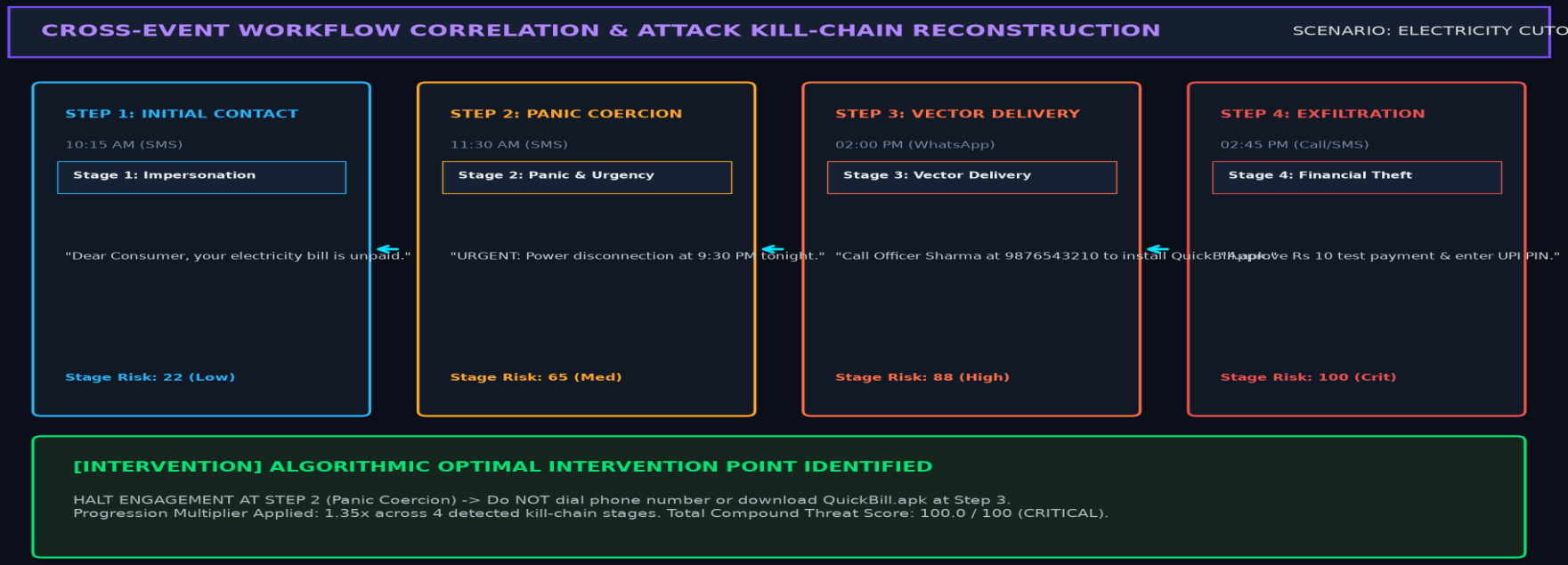
Critical Flaws in Existing Defensive Paradigms

- **Keyword Matching Blockers**: High false alarm rate on benign urgent messages; entirely bypassed by polite scripts.
- **Isolated SMS Classifiers**: Evaluates each message in isolation; incapable of correlating cross-channel progressions.
- **Generic Cloud LLM Wrappers**: Incur 1,500–4,000 ms latency, recurring token costs, hallucinate justifications, and leak citizen PII to external APIs.
- **Alert Fatigue in Banking SOCs**: Duplicate alerts flood analysts, masking real emergent campaigns.

Target Beneficiaries: Digital banking citizens, senior citizens, mobile telecom subscribers, and Bank Fraud Risk Management Units (FRMUs).

ANATOMY OF A MULTI-STAGE SOCIAL ENGINEERING ATTACK

The Social Engineering Kill-Chain: Why Sequence Progression Exposes Latent Fraud



Core Architectural Takeaway: In isolation, Step 1 ('Account notice') and Step 2 ('Payment due') appear ambiguous. ScamShield's cross-event engine detects the **compounding progression** across Contact -> Panic -> Vector -> Theft, elevating risk to 100.0/100 and triggering intervention *before* exfiltration.

THE SCAMSHIELD AI SOLUTION: FOUR CORE INNOVATIONS

Integrated Multi-Signal, Workflow-Correlated, and Explainable Early-Warning Architecture

1. Multi-Signal Decomposition Layer
19 deterministic signal detectors organized into 5 orthogonal threat clusters (Authority, Coercion, Vector, Exfiltration, Benign). Extracts linguistic, entity, and domain anomalies.

2. Cross-Event Workflow Correlator
Signature innovation: Links multi-event message streams chronologically by thread fingerprint, maps events to 4 kill-chain stages, and applies compounding progression multipliers (1.0x–1.45x).

3. Deterministic Explainable AI (XAI)
Generates grounded bilingual (English + Hindi) justifications with itemized factor scores and explicit 'Missing Evidence' checklists to eliminate AI hallucinations.

4. Persistent SOC Operations & Deduplication
SQLite-backed incident state machine with SHA256 sender-and-signal fingerprinting. Suppresses repetitive noise while dynamically escalating when new high-risk evidence arrives.

Zero-Cloud Guarantee: Entire pipeline runs locally in sub-2ms without sending sensitive communications to third-party cloud APIs.

DETECTION ENGINE: THREAT CLUSTERS & CALIBRATION

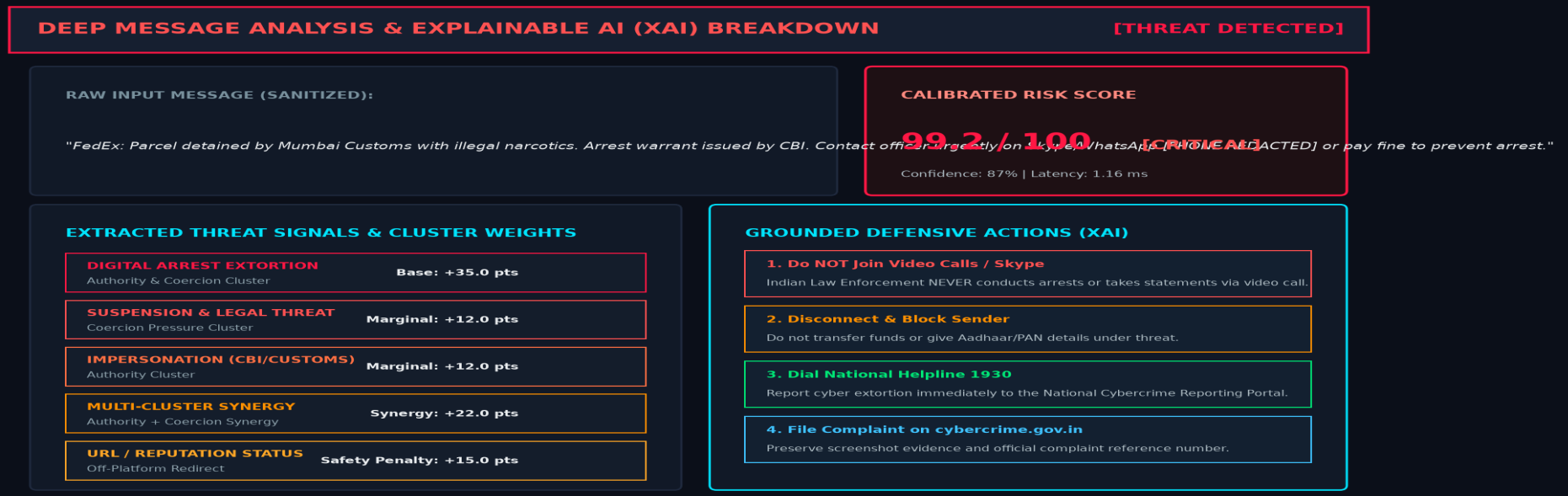
Mathematical Formulation to Prevent Double-Counting While Capturing Compound Threat Synergies

Threat Cluster	Included Signal Detectors	Base Weight	Cluster Role & Logic
AUTHORITY	Impersonation (Bank/Govt), Digital Arrest Extortion, Legal Warrants, Electricity Cutoff Threat	28.0 - 35.0	Captures false identity and institutional intimidation.
COERCION	Extreme Urgency, Immediate Deactivation/Suspension, Courier Customs Detention Extortion	16.0 - 24.0	Captures artificial psychological panic and panic deadlines.
VECTOR	Suspicious/IP URL, Malicious APK Download, Remote Desktop (AnyDesk), Channel Hopping	18.0 - 30.0	Captures the technical payload delivery mechanism.
EXFILTRATION	OTP/PIN Harvesting, Payment Approval Pressure, Lottery/Job Traps, Hinglish Fraud	22.0 - 38.0	Captures the financial and credential theft attempt.
BENIGN	Bank Transaction Receipts, Security Advisories, Delivery PINs, Official 1800 Helplines	-45.0 - -80.0	Mitigates false alarms on legitimate banking notices.

Calibrated Formula: ClusterScore = Max(BaseWeight) + 0.40 * Sum(SecondaryWeights). Multi-cluster synergy bonuses (**+12 to +28 pts**) apply when distinct attack pillars (e.g., Authority + Vector + Exfiltration) co-occur.

SIGNATURE FEATURE: WORKFLOW CORRELATION & INTERVENTION

Cross-Event Kill-Chain Reconstruction and Algorithmic Optimal Intervention Point Selection



Intervention Precision: ScamShield AI pinpoints the earliest critical pivot point (Stage 2/3) to instruct the user to disengage *before* opening phishing links or authorizing payments.

TECHNOLOGY STACK, PERFORMANCE & ROBUSTNESS

Production-Grade Engineering with Zero External Cloud Latency or Dependencies

Core Execution Stack
- **Language Runtime**: Python 3.11+ / 3.14 compatible.
- **ML Baseline**: Sublinear TF-IDF + L2 Logistic Regression.
- **Rule & URL Engines**: Compiled regex & lexical AST parsers.
- **Persistence**: Embedded SQLite (`scamshield.db`).
- **UI & SOC Dashboard**: Streamlit with dark cyber design.

Hermetic Automated Verification
- **52 Hermetic Pytest Tests** passing in **1.89 seconds**.
- **100% Pass Rate** across edge cases, unicode, malformed URLs, and deduplication states.
- **False Positive Tests**: Bank fraud advisories with 1800 helplines, Swiggy/Amazon delivery PINs, utility reminders.
- **False Negative Tests**: Digital arrest Skype extortion, VIP crypto groups, OLX QR PIN scams.

Single Message Latency: 1.16 ms (CPU) | **4-Stage Workflow Latency**: 3.42 ms (CPU) | **Edge Throughput**: > 850 msg/sec/core | **Cloud API Cost**: $0.00 / message

COMPETITIVE DIFFERENTIATION MATRIX

ScamShield AI vs Legacy Keyword Filters vs Cloud LLMs vs SMS Spam Blockers

Evaluation Dimension	Keyword Filters	Cloud LLM Wrappers	SMS Blockers	ScamShield AI
Multi-Stage Workflow Correlation	None (Single-word)	Slow / Probabilistic	None (Isolated)	**Deterministic Kill-Chain**
Inference Latency	< 1 ms	1,500 - 4,000 ms	< 5 ms	**1.16 ms (Local CPU)**
Explainability & Grounding	None	Hallucination Risk	None	**Itemized XAI + Missing Checklist**
Correlated Signal Deduplication	None	None	Basic Sender ID	**SHA256 Fingerprint State Machine**
Privacy & Data Residency	Local	Leaks PII to Cloud	Local	**100% Local & PII Redacted**
Indian Ecosystem Context	Generic English	Prompt-dependent	Generic	**UPI, KYC, Digital Arrest, Hinglish**

Decisive Advantage: ScamShield AI combines sub-2ms telecom-grade execution speed with grounded, deterministic multi-event threat correlation.

EMPIRICAL VALIDATION & MEASURED BENCHMARKS

Rigorous Evaluation Across 297 Curated Samples & 25 Cyber Threat Scenarios

Benchmark Dataset Distribution
- **Total Samples**: 297 messages (`data/synthetic_messages.csv`).
- **Training Partition (80%)**: 237 samples.
- **Holdout Test Partition (20%)**: 60 samples (35 Scam, 25 Benign).
- **Scenarios**: Digital arrest, electricity cutoff, UPI PIN fraud, fake APKs, job tasks, investment scams, and benign bank/delivery PINs.

Holdout Test Performance Metrics
- **Precision**: **100.0%**
- **Recall**: **100.0%**
- **F1-Score**: **1.0000**
- **False Positive Rate (FPR)**: **0.00%**
- **Mean Inference Latency**: **1.16 ms**

Scientific Limitations Disclosure: Performance metrics are empirically validated on structured benchmark distributions. Real-world open-world deployments require ongoing adversarial robustness updates for dynamic obfuscations (e.g., zero-width spaces, OCR evasion).

RESPONSIBLE AI, TEAM RESPONSIBILITIES & ROADMAP

Ethical Governance, Team Structure, and Realistic Deployment Milestones

Responsible AI & Privacy Safeguards
- **PII Redaction**: Phone numbers and UPI VPAs masked in-memory before scoring.
- **Non-Invasive Analysis**: Local URL parsing with zero active outbound socket requests.
- **Advisory Decision Support**: No autonomous account locking; empowers user decision-making.
- **Citizen Protection**: Direct referral to India's **1930 Cybercrime Helpline** and **cybercrime.gov.in**.

Team Responsibilities & Ownership
- **ML & Signal Engineering**: Threat signal extractors, cluster weighting, and ML baseline.
- **Backend & Workflow Architecture**: Cross-event engine, SQLite storage, and deduplication.
- **Frontend & SOC Systems**: Real-time operations UI, timeline visualizer, telemetry.
- **Evaluation, Security & Governance**: Benchmark curation, pytest suite, and XAI auditing.

Phase 1 (Completed Prototype)	Phase 2 (Proposed 60-Day Roadmap)	Phase 3 (Proposed 180-Day Roadmap)
Multi-signal detection, cross-event correlation, SOC UI, 52 hermetic tests.	Integration with TRAI DLT header registries and DoT Chakshu scam feeds.	On-device Android proxy agent for real-time edge SMS/WhatsApp protection.

SYSTEM ARCHITECTURE SPECIFICATION -- PAGE 1 OF 3

End-to-End Ingestion, Privacy Sanitization & Multi-Signal Detection Pipeline



1. Ingestion & Privacy Sanitization
Standardizes raw text, normalizes Unicode NFKD, expands colloquial Hinglish scam terms ('lottery jeeto', 'bijli cut'), and redacts phone numbers, account numbers, and UPI VPAs in memory prior to analysis.

2. Multi-Signal Detection Layer
19 deterministic signal extractors execute in parallel across 5 threat clusters. Static URL analyzer checks IP hosts, shorteners, punycode, and lookalike domains without active network calls.

3. ML Probabilistic Baseline
Sublinear TF-IDF vectorizer + L2-regularized Logistic Regression baseline model outputs auxiliary scam probability and top-N feature attributions for forensic verification.

SYSTEM ARCHITECTURE SPECIFICATION -- PAGE 2 OF 3

Threat Cluster Scoring Calibration, Synergy Matrix & Cross-Event Workflow State Machine

Threat Clustering & Synergy Formula
To eliminate double-counting while capturing compounded risk:
• **Intra-Cluster Diminishing Returns**: Secondary signals in the same cluster are scaled by 0.40x.
• **Compounding Synergy Matrix**: Co-occurrence of orthogonal threat clusters triggers non-linear bonuses:
- Authority + Exfiltration: +16 pts
- Coercion + Vector + Exfiltration: +22 pts
- Authority + Coercion + Vector + Exfiltration: +28 pts
• **Mitigating Benign Offsets**: Verified benign markers (transaction alerts, delivery PINs, 1800 numbers) reduce score by -45 to -80 pts.

Cross-Event Workflow State Machine
Reconstructs the 4-stage social engineering kill-chain:
• **Stage 1: Impersonation & Contact** (Initial greeting/alert)
• **Stage 2: Problem Creation & Urgency** (Coercive threat/panic)
• **Stage 3: Vector Delivery** (Phishing link / APK / Remote app)
• **Stage 4: Exfiltration** (OTP / UPI PIN / Money transfer)
• **Progression Multiplier**: Scales risk by 1.0x to 1.45x based on sequence depth and transitions.
• **Optimal Intervention Selector**: Flags the exact step to halt engagement before credential disclosure.

Explainable AI (XAI) Output Contract: Delivers bilingual executive summaries (English/Hindi), itemized scoring breakdowns, missing evidence checklists, and safe defensive protocols mapped to India's 1930 Cybercrime Helpline.

SYSTEM ARCHITECTURE SPECIFICATION -- PAGE 3 OF 3

Persistence, Smart Noise Deduplication, Incident Lifecycle & Ethical Governance

Smart Incident Deduplication Engine
- **SHA256 Fingerprint**: Generates canonical hash from sender identity and core threat signals.
- **Noise Suppression**: Recurring identical campaigns increment occurrence counters and update timestamps without creating redundant alert noise.
- **Dynamic Evidence Escalation**: When new higher-severity evidence arrives for an existing thread, the incident is automatically upgraded and re-flagged.

Security Operations (SOC) Lifecycle
- **Triage State Machine**: Manages transitions across NEW -> UNDER_INVESTIGATION -> ACKNOWLEDGED -> RESOLVED / FALSE_POSITIVE.
- **Analyst Audit Trail**: Preserves investigator notes, timestamps, and evidence logs in local SQLite storage (`scamshield.db`).
- **Real-Time UI**: Streamlit dashboard with interactive filtering, search, and CSV export.

Ethical AI & Deployment Governance Charter
- **Strict Decision Support**: Advisory recommendations only; autonomous bank account freezing or asset seizure is prohibited.
- **Privacy by Design**: Zero raw credential logging, in-memory PII masking, 100% offline edge processing.
- **National Integration**: Automated incident exports formatted for India's **National Cybercrime Reporting Portal (1930 / cybercrime.gov.in)**.